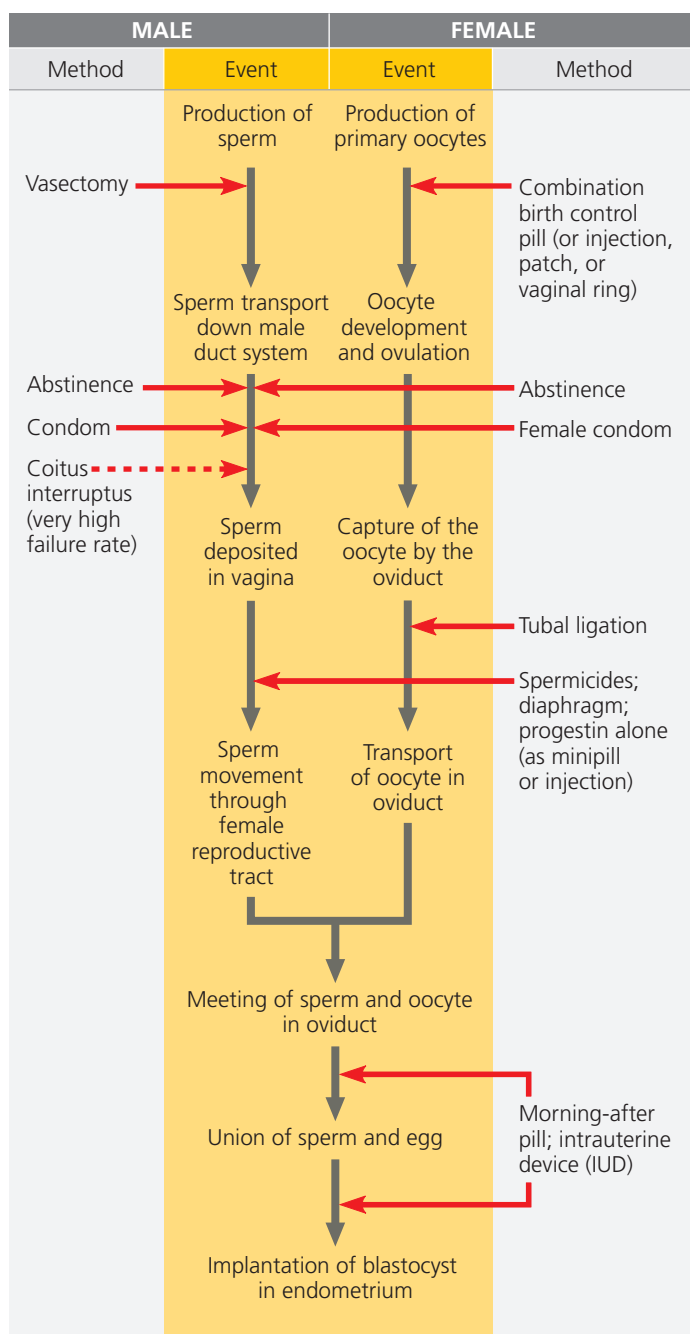


▼ **Figure 46.20 Mechanisms of several contraceptive methods.** Red arrows indicate where these methods, devices, or products interfere with events from the production of sperm and primary oocytes to implantation of a developing embryo.



As a method of preventing fertilization, *coitus interruptus*, or withdrawal (removal of the penis from the vagina before ejaculation), is unreliable. Sperm from a previous ejaculate may be transferred in secretions that precede ejaculation. Furthermore, a split-second lapse in timing or willpower can result in tens of millions of sperm being transferred before withdrawal.

Used properly, several methods of contraception that block sperm from meeting the egg have pregnancy rates of less than 10%. The *condom* is a thin, latex rubber or natural membrane sheath that fits over the penis to collect the semen. For sexually active individuals, latex condoms are the only contraceptives that are highly effective in preventing the spread of AIDS and other *sexually transmitted infections (STIs)*, also known as *sexually transmitted diseases (STDs)*. This protection is not absolute, however. Another common barrier device is the *diaphragm*, a dome-shaped rubber cap inserted into the upper portion of the vagina before intercourse. Both of these devices have lower pregnancy rates when used in conjunction with a spermicidal (sperm-killing) foam or jelly. Other barrier devices include the vaginal pouch, also called a “female condom.”

Except for complete abstinence from sexual intercourse or sterilization (discussed later), the most effective means of birth control are the intrauterine device (IUD) and hormonal contraceptives. The IUD has a pregnancy rate of 1% or less and is the most commonly used reversible method of birth control outside the United States. Placed in the uterus by a doctor, the IUD interferes with fertilization and implantation. Hormonal contraceptives, most often in the form of **birth control pills**, also have pregnancy rates of 1% or less.

The most commonly prescribed hormonal contraceptives contain a synthetic estrogen and a synthetic progesterone-like hormone called progestin. This combination mimics negative feedback in the ovarian cycle, stopping the release of GnRH by the hypothalamus and thus of FSH and LH by the pituitary. The prevention of LH release blocks ovulation. In addition, the inhibition of FSH secretion by the low dose of estrogens in the pills prevents follicles from developing.

Another hormonal contraceptive with a very low pregnancy rate contains only progestin. Progestin causes thickening of a woman’s cervical mucus so that it blocks sperm from entering the uterus. Progestin also decreases the frequency of ovulation and causes changes in the endometrium that may interfere with implantation if fertilization occurs. This contraceptive can be administered as injections that last for three months or as a tablet (“minipill”) taken daily.

Hormonal contraceptives have both harmful and beneficial side effects. They increase the risk of some cardiovascular disorders slightly for nonsmokers and quite substantially (three- to tenfold) for women who smoke regularly. At the same time, oral contraceptives eliminate

couple needs knowledge of physiological indicators associated with ovulation, such as changes in cervical mucus. Note also that a pregnancy rate of 10–20% is typical for couples practicing natural family planning. (In this context, pregnancy rate is the percentage of women who become pregnant in one year while using a particular pregnancy prevention method.)